Unknown A: 2-Bromobutane

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30 November 2023

Chem 2081-007

Introduction

The goal of this lab report is to identify an unknown compound using nuclear magnetic resonance (NMR), gas chromatography/mass spectroscopy (GCMS), and infrared spectroscopy (FTIR). Understanding how to apply these methods of chemical analysis is very important for being able to confirm the product of a reaction.

Body

As mentioned, the three main methods of chemical analysis are NMR, GCMS, and FTIR. NMR uses electromagnetic radiation to excite the nuclei of all the ^{1}H or ^{13}C in a compound. As the nuclei return to their stable energy states, they emit frequencies which are detected by the NMR instrumentation. The signals produced by this are known as free-induction decay, or FID. Fourier transform mathematics are used to transform these signals into a normal NMR spectrum (Keller, 2023). In an ^{1}H NMR spectra, the number of groups of signals correlates to the number of different kinds of protons in a compound. The integration gives the relative quantities of the number of protons in each group. The horizontal axis of an NMR spectra shows the chemical shift (ppm), which provides the most useful information because it directly relates to the molecule's structure. Lastly, the coupling patterns of signals provide information about neighboring protons (Keller, 2023).

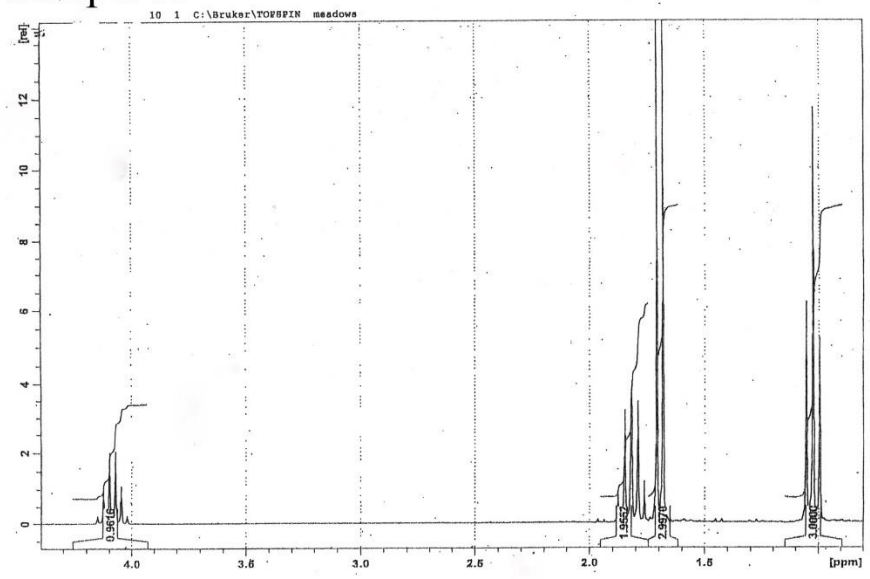
In mass spectroscopy, an ion source converts a compound into a gas-phase ion. A mass analyzer then sorts the ions by their mass-charge (m/z) ratios. Then, a detector records the number of ions of each m/z value, creating a mass spectrum (Keller, 2023). GCMS uses a combined gas chromatograph and mass spectrometer, where samples of the emitted stream from the gas chromatograph are directed into the mass spectrometer, which then undergo ionization and mass analysis. Mass spectral data is presented in a graph of relative abundance vs m/z

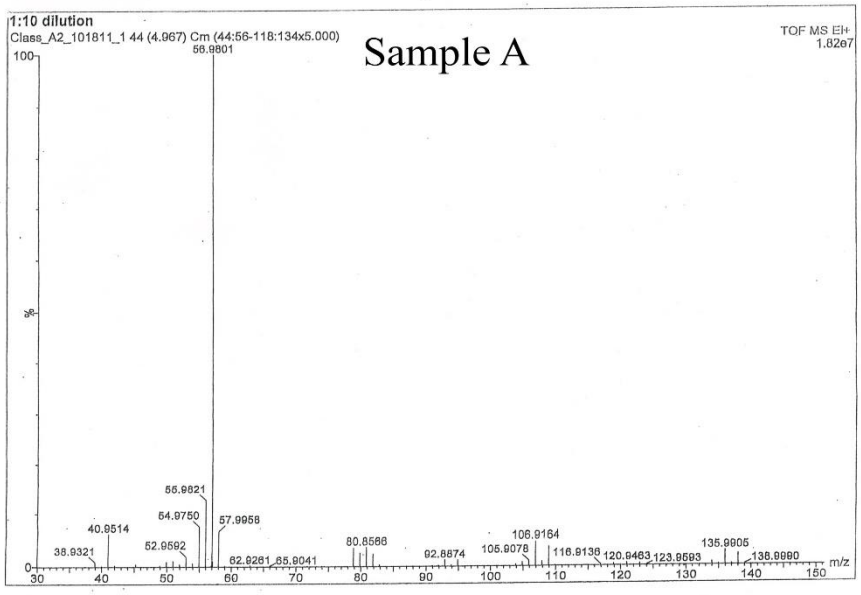
values. The most intense peak of the spectrum is referred to as the base peak, which comes from the most stable fragment. The tallest peak of the last cluster is the molecular ion peak, which tells you the molar mass of the compound. If the molecular ion peak is odd, the molecules must contain an odd number of nitrogen atoms. On the other hand, if the value of the peak is even, there must be either no nitrogen atoms or an even number of nitrogen atoms. The (M+1) and (M+2) peaks appear in the presence of isotopes and can provide information about the molecular formula of the compound. If there is an (M+2) peak, that suggests that there is a bromine or chlorine in the molecule, as those elements have two abundant isotopes (Keller, 2023).

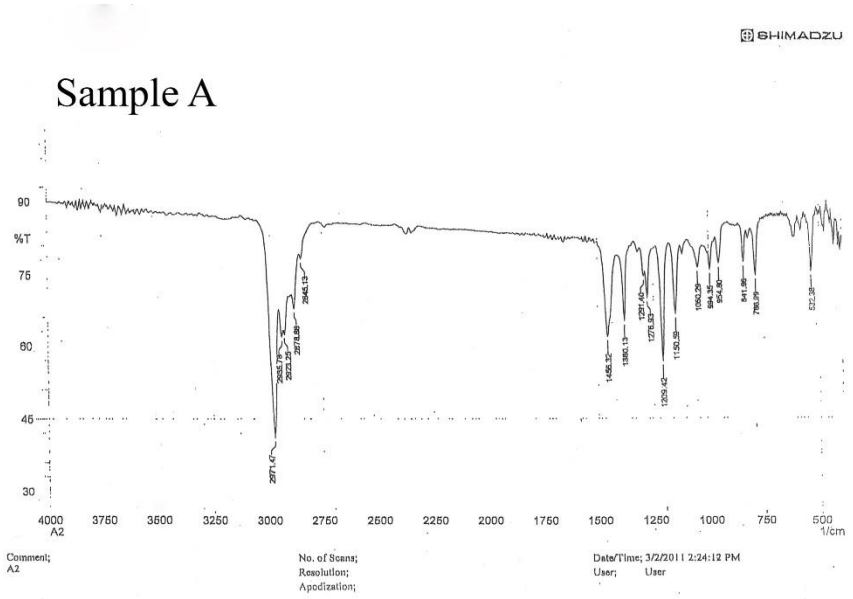
In FTIR, infrared energy is passed through a sample, and an absorption band appears in an IR spectrum at the frequency in which molecular vibration occurs in the molecule. The IR spectrum can be used to identify specific functional groups in the molecule. There are three regions in an IR spectrum: functional, fingerprint, and aromatic. The functional group region (1500-4000 cm^{-1}) provides information for most major functional groups, and this is the region that is primarily used when trying to identify compounds. The fingerprint region (900-1500 cm^{-1}) is more complex and usually displays a greater concentration of peaks. The aromatic region (600-900 cm^{-1}) provides information about alkenes and aromatic compounds (Keller, 2023).

Data

Sample A


10 1 C:\Bruker\TOPSPIN meadows
[rel]
0.9616
1.9552
2.9570
3.0000
4.0
3.5
3.0
2.5
2.0
1.5
[ppm]


1:10 dilution
Class_A2_101811_1 44 (4.967) Cm (44:56-118:134x5.000)
Sample A
TOF MS EI+
1.82e7
100
%
56.9801
55.9821
54.9750
57.9958
40.9514
52.9592
38.9321
80.8566
62.8874
106.9164
105.9078
116.9136
120.9463
123.9593
135.9905
138.9980
m/z



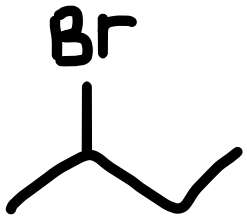
Analysis

In the mass spectrum, the molecular ion region falls in the 135-140 m/z range. There also appears to be a (M+2) peak, suggesting there are two abundant isotopes. The second peak is close to the height of the molecular ion peak, which is characteristic of a bromine atom. Looking at the IR spectrum, the peaks in the diagnostic region lay in the 2600-3000 cm^{-1} range. The bonds that typically appear in this region are C-H bonds. This IR spectrum does not show any peaks in the double or triple bond regions, nor any broad peaks in the 3000+ region which would signify the presence of O-H or N-H bonds. Knowing that this molecule contained only C-H and C-C bonds, as well as at least one bromine atom, the ^{1}H NMR spectrum was analyzed. It showed a 3H triplet, a 3H doublet, a 2H quintet, and a 1H sextet. Since there are two groups with 3H, it was determined that there were two methyl groups. The 3H triplet was around 1.0 ppm, and the 3H doublet was around 1.7 ppm. Therefore, the 3H doublet is closer to the bromine atom because the electronegativity of the bromine deshields the protons and moves it further downfield, giving it a higher ppm. The 1 H sextet was around 4.1 ppm, which means that this

hydrogen is likely part of a secondary alkyl halide. This is backed up by the fact that the neighboring methyl group is a doublet, so it only has one hydrogen on the adjacent carbon. That leaves the 2H quintet which was around 1.85 ppm. This means it is a methylene group with 4 neighboring hydrogens, so it must be between the 3H triplet and the 1H sextet.

Conclusion

Combining all this information, the molecule was determined to be 2-bromobutane. The molecular weight of 2-bromobutane is 137.02, which matches with the molecular ion region of the mass spectrum.



References

Keller L, Lichter J, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Laboratory Manual [lab manual]. Auburn (Al): Auburn University.